

PROJECT SYNOPSIS

Title of the Project

SmartGrid Guardian: AI-Powered Smart Grid Stability Prediction Using Machine Learning

1. Introduction

Provide background information about smart grids, the importance of grid stability, and the role of machine learning in modern power systems.

2. Problem Statement

Power grids face challenges such as fluctuating demand, delayed system responses, and power imbalances that may lead to instability. Traditional monitoring systems may not always provide predictive insights. Therefore, an intelligent system is required to predict grid stability before critical failures occur.

3. Objectives

- Predict smart grid stability using machine learning.
 - Compare multiple machine learning algorithms.
 - Identify the most accurate prediction model.
 - Develop an interactive dashboard for stability analysis.
 - Support decision-making through data-driven insights.
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4. Literature Review (Optional but Recommended)

Briefly discuss:

- Smart Grid Technology
- Machine Learning in Power Systems
- Grid Stability Prediction Techniques
- Explainable AI in Energy Systems

5. Proposed Methodology

Dataset

Smart Grid Stability Dataset

Data Preprocessing

- Data cleaning
- Feature selection
- Data normalization

Machine Learning Models

- Logistic Regression
- Decision Tree
- Random Forest
- XGBoost

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-Score

Dashboard Development

- Streamlit-based user interface
 - Stability prediction visualization
 - Risk assessment display
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6. SDG Mapping

SDG 7 – Affordable and Clean Energy

Supports efficient and reliable energy management.

SDG 9 – Industry, Innovation and Infrastructure

Promotes intelligent infrastructure through AI-based monitoring.

SDG 11 – Sustainable Cities and Communities

Contributes to reliable electricity systems for smart cities.

7. Expected Outcomes

- Accurate grid stability prediction.
 - Identification of important influencing parameters.
 - Interactive dashboard for monitoring predictions.
 - Demonstration of AI applications in power systems.
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8. Tools and Technologies

Programming Language:

- Python

Libraries:

- Pandas
- NumPy
- Scikit-Learn
- XGBoost
- Plotly

Framework:

- Streamlit

Version Control:

- Git & GitHub
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9. Conclusion

The proposed system aims to leverage machine learning techniques for predicting smart grid stability and providing a practical decision-support tool through an interactive dashboard.

References

<https://www.kaggle.com/datasets/pcbreviglieri/smart-grid-stability>.